Chapter 11, Problem 74P

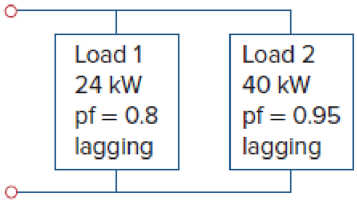
Problem

A 120-V rms 60-Hz source supplies two loads connected in parallel, as shown in Fig. 11.89.

(a) Find the power factor of the parallel combination.

(b) Calculate the value of the capacitance connected in parallel that will raise the power factor to unity.

Figure 11.89:



Step-by-step solution

Step 1 of 4

(a)

$$\theta_1 = \cos^{-1}(0.8)$$

$$\theta_1 = 36.87°$$

$$S_1 = \frac{P_1}{\cos\theta_1}$$

$$S_1 = \frac{24\times10^3}{0.8}$$

$$S_1 = 30\text{k VA}$$

$$Q_1 = S_1 \sin\theta_1$$

$$Q_1 = (30\times10^3)(0.6)$$

$$Q_1 = 18\text{k VAR}$$

$$S_1 = 24 + j18\text{k VA}$$

Step 2 of 4

$$S_2 = \frac{P_2}{\cos\theta_2}$$

$$S_2 = \frac{40\times10^3}{0.95}$$

$$S_2 = 42.105\text{k VA}$$

$$Q_2 = S_2 \sin\theta_2$$

$$Q_2 = 13.144\text{k VAR}$$

$$S_2 = 40 + j13.144\text{k VA}$$

Step 3 of 4

$S = S_1 + S_2$

$S = 64 + j31.144 \text{ kVA}$

$\theta = \tan^{-1}\left(\frac{31.144}{64}\right)$

$\theta = 25.95°$

$\text{Pf} = \cos\theta$

$\boxed{\text{Pf} = 0.899}$ [Lagging]

Step 4 of 4

(b)

$\theta_2 = 25.95°$ and

$\theta_1 = 0$

$Q_C = P(\tan\theta_2 - \tan\theta_1)$

$Q_C = 64 \times 10^3 [\tan 25.95° - 0]$

$Q_C = 31.144 \text{ k VAR}$

$C = \frac{Q_C}{\omega V_{rms}^2}$

$C = \frac{31.144 \times 10^3}{2\pi \times 60 \times (120)^2}$

$\boxed{C = 5.74 \text{ mF}}$